

WIRELESS SECURITY STANDARDS

UNIT 7



WIRELESS SECURITY STANDARDS

- Wireless security standards are protocols designed to protect wireless networks (Wi-Fi) from unauthorized access, data breaches, and interference.

1. **Wired Equivalent Privacy (WEP)**

- Introduced in 1997 as part of the original Institute of **Electrical and Electronics Engineers (IEEE) 802.11** standard.
- The goal was to provide privacy similar to that of wired networks.
- It uses **Rivest Cipher 4 (RC4)** stream cipher for encryption and an **Initialization Vector (IV)** 24-bit field added to it.
- Key length 64-bit (40-bit key + 24-bit IV) or 128-bit (104-bit key + 24-bit IV).
- Authentication methods include **Open System** (any device can authenticate) and **Shared Key** (devices authenticate using a shared secret)

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WIRELESS SECURITY STANDARDS (CONTINUES…)

Weaknesses of WEP

i. IVs are too short and frequently reused

ii. RC4 is vulnerable to statistical attacks

iii. Key reuse makes WEP vulnerable to replay attacks

Real-World Attack

- Tools like Aircrack-ng can capture encrypted packets and use them to recover WEP keys.
- Declared insecure by the Wi-Fi Alliance in 2004.

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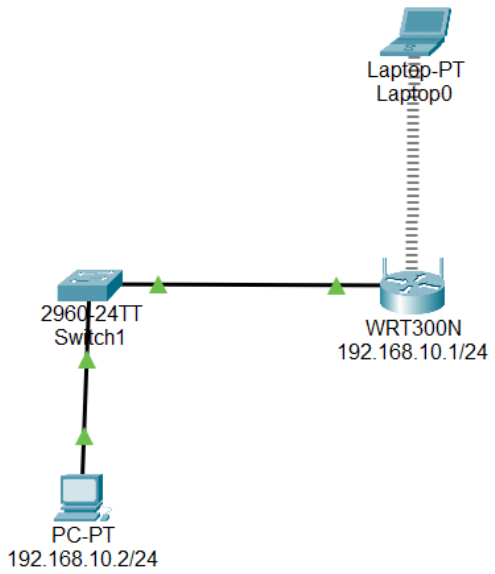
PRACTICAL LAB

WEP

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WEP

NETWORK TOPOLOGY



OBJECTIVES

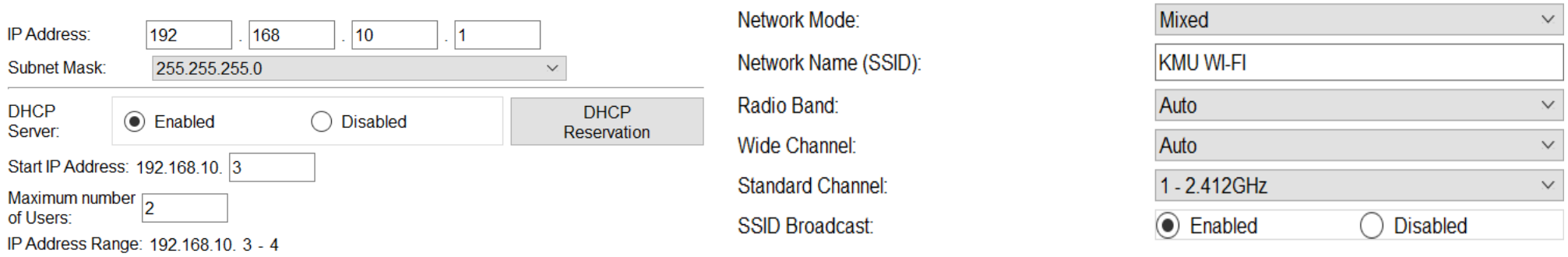
1. Configuring the network with IP Addresses.
2. Setting up a WEP 64-bit encryption key.
3. Establishing a successful WEP connection.

TASK: Set up the same network topology and configure WEP with a 128-bit encryption key and an Access Point (AP) in place of the Wireless Router.

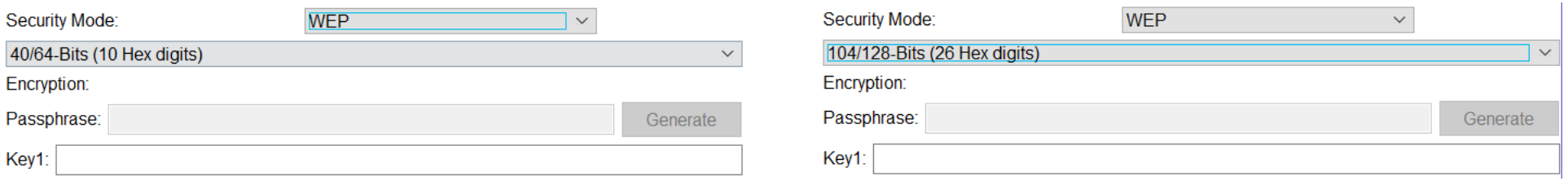
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WIRELESS ROUTER CONFIGURATION

- Let's configure the Wireless Router with the aim of connecting only one device (Laptop0).



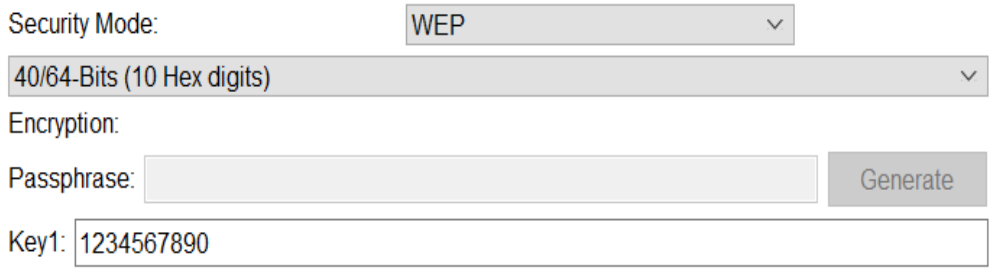
- **NOTE:** There are two encryptions keys we can use to encrypt data, which are;



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WEP KEY SET UP

- These keys are **hex**, meaning we are dealing with base 16 which are 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E and F.



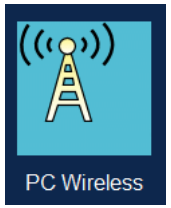
- Let's make use that the Laptop is ready to connect to the Wireless Router.



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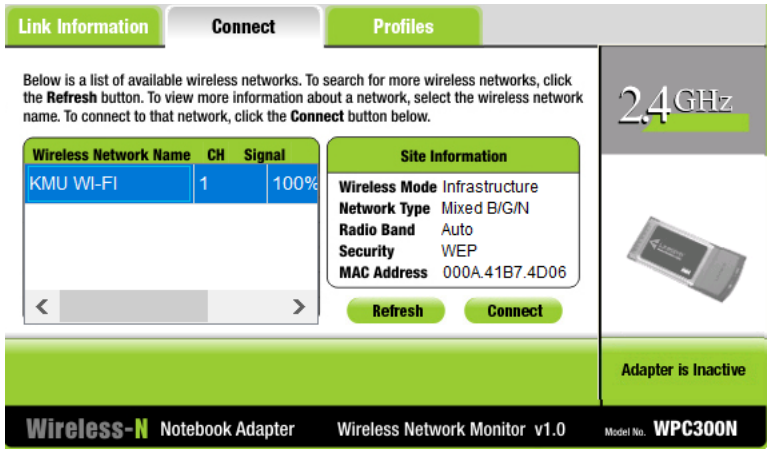
WEP CONNECTION

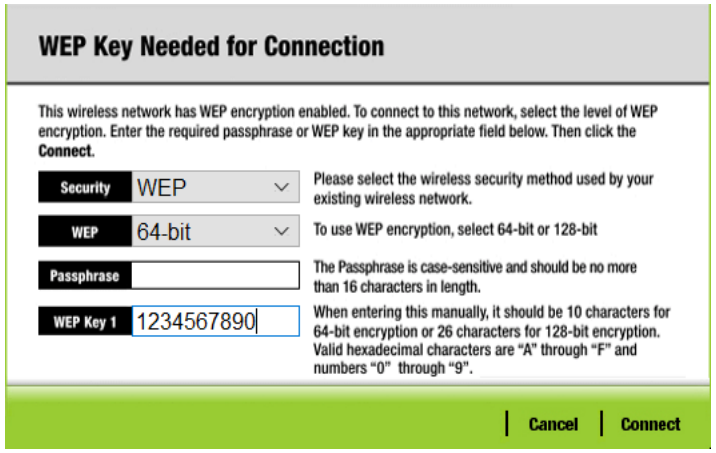
- Now it's time that our Laptop get's connected to the Wireless Router via WEP with the Shared Key.

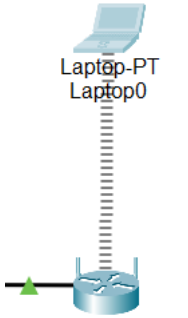




Laptop-PT
Laptop0







Laptop-PT
Laptop0

WRT300N
192.168.10.1/24



WRT300N
192.168.10.1/24

DHCP	Static
IPv4 Address	192.168.10.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

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WIRELESS SECURITY STANDARDS (CONTINUES…)

2. **Wi-Fi Protected Access (WPA)**

- Introduced in 2003 by the **Wi-Fi Alliance**.
- Interim solution to replace WEP until IEEE 802.11i was finalized.
- It uses a **Temporal Key Integrity Protocol (TKIP)** which dynamically generates new keys for every packet and appends a **Message Integrity Check (MIC)** to detect tampering.
- Still uses RC4 for encryption.

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WIRELESS SECURITY STANDARDS (CONTINUES…)

WPA modes

i. **WPA-Pre-Shared Key (PSK)** – For home users.

ii. **WPA-Enterprise** – Uses IEEE 802.1X authentication and a Remote Authentication Dial-In User Service (RADIUS) server.

Weaknesses & Real-World Attack

i. TKIP still relies on RC4, inheriting many of its weaknesses

ii. Vulnerable to replay attacks and MIC key recovery

- **Beck-Tews** attack allows packet injection by exploiting TKIP flaws

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WIRELESS SECURITY STANDARDS (CONTINUES…)

3. **Wi-Fi Protected Access II (WPA2)**

- Released in 2004, based on the **IEEE 802.11i** amendment.
- Made mandatory for Wi-Fi certified devices by 2006.
- It used **Advanced Encryption Standard (AES)** and **Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP).**
- Provides both **encryption** and **message integrity**.
- Uses a **4-way handshake** to establish encryption keys.
- Authentication Modes include **WPA2-PSK** and **WPA2-Enterprise**.
- **Key Reinstallation Attack (KRACK**) vulnerability of 2017 exploits flaws in the 4-way handshake process to reinstall already-used keys.

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WIRELESS SECURITY STANDARDS (CONTINUES…)

4. **Wi-Fi Protected Access III (WPA3)**

- Introduced in 2018 to overcome WPA2 limitations.
- Replaces WPA2's PSK with **Simultaneous Authentication of Equals (SAE)** which is based on **Password-Authenticated Key Exchange (PAKE),** providing resistance to dictionary attacks.
- Past session keys remain secure even if a current key is compromised (**Forward Secrecy**).

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WIRELESS SECURITY STANDARDS (CONTINUES…)

Additional Features

i. **Opportunistic Wireless Encryption (OWE)** – Provides encrypted communication in open networks

ii. **Wi-Fi Easy Connect** – Uses **Device Provisioning Protocol (DPP)** to connect IoT (Internet of Things) devices securely via QR codes

Vulnerability

i. **Dragonblood attacks (2019)** – Exploited implementation flaws in the SAE protocol, now mitigated in updates

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PRACTICAL LAB

WPA

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WPA

WPA PERSONAL

- For this lab, utilize the previous Network Topology and follow the guide lines on the next page.

TASK: Configure WPA Personal with AES encryption and establish a successful connection. What do you think, is AES also supported in WPA I or just WPA II?

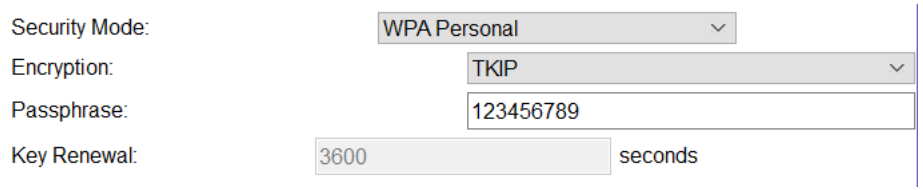
WPA ENTERPRISE TASK

- Configure WPA2 Enterprise using the same enterprise topology given.

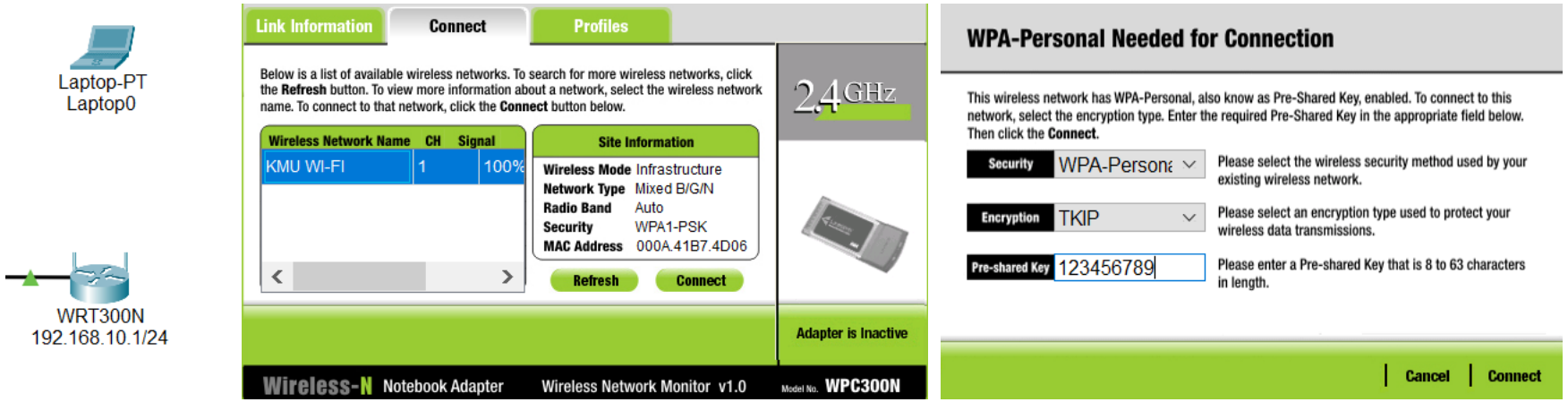
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WPA PERSONAL

- Set up the WPA Personal on the Wireless Router as shown below;



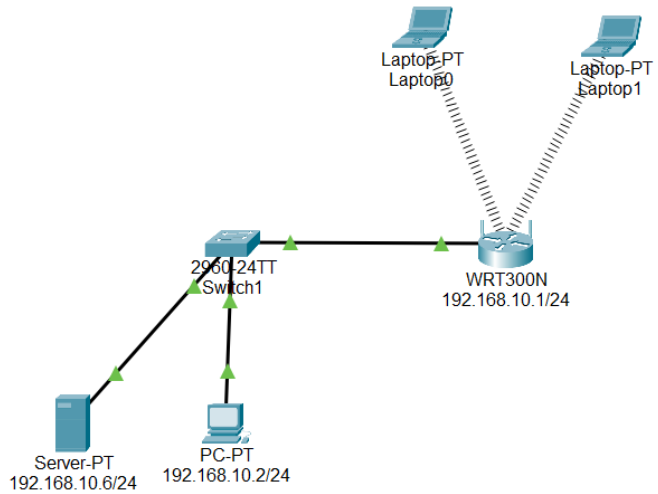
- Now let's connect the Laptop to the Wireless Router via WPA Personal (TKIP).



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WPA ENTERPRISE

NETWORK TOPOLOGY



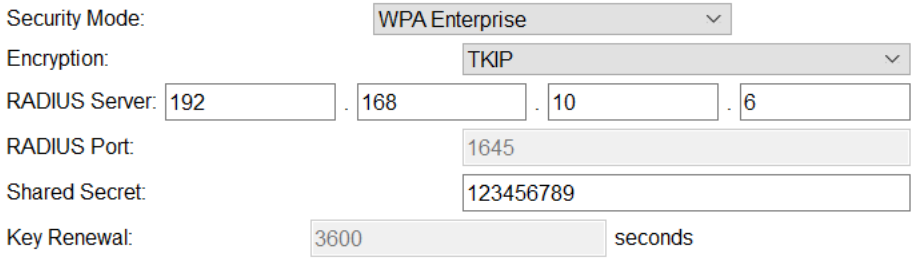
OBJECTIVES

1. Re-configure the Wireless Router DHCP to accommodate another device with a range 3 – 5.
2. Configuring WPA Enterprise on the Wireless Router.
3. Setting up Client WPA Enterprise credentials.
4. Setting up both the WPA Enterprise Client and Wireless credentials on the RADIUS Server.
5. Establishing a successful WPA Enterprise connection.

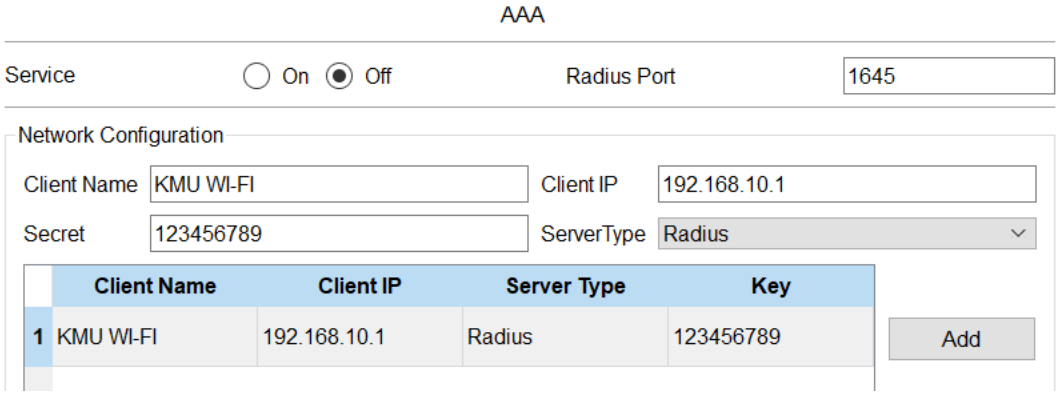
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SETTING UP WPA ENTERPRISE

- On this one with switch up things, we need a Server to tag along, and another device for this lab.
- Let's configure the Wireless Router with WPA Enterprise.



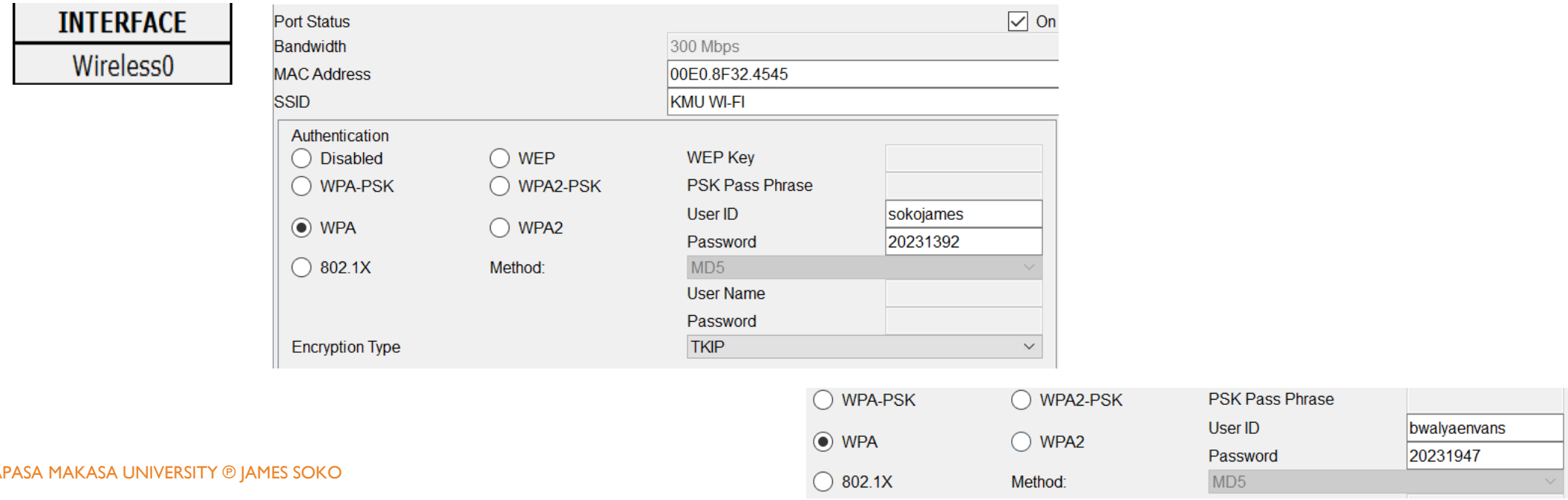
- Let's start off by configuring the Server with the AAA service.



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SETTING UP CLIENT WPA ENTERPRISE CREDENTIALS

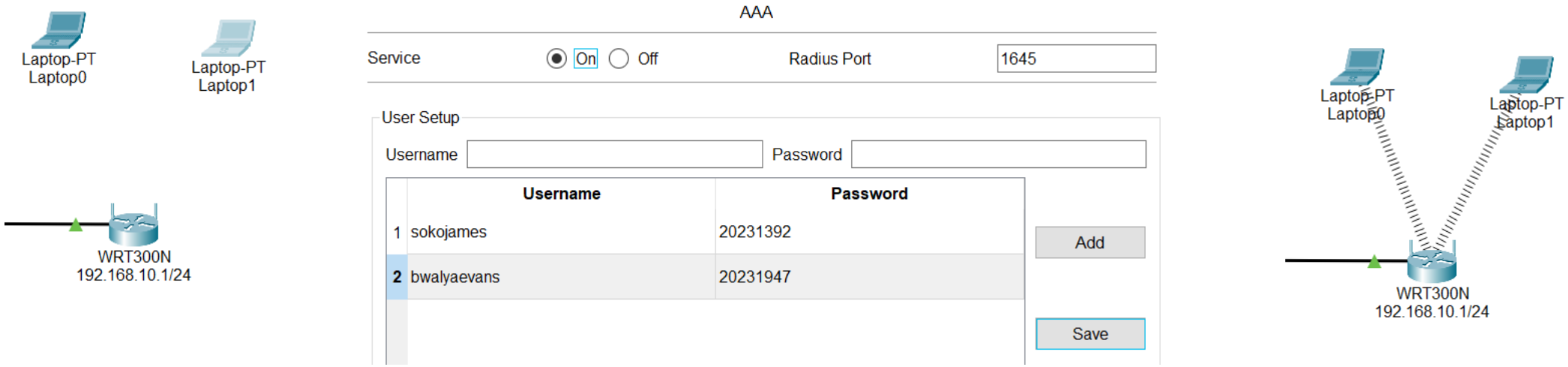
- Let's assume this is an organization, such as Kapasa Makasa University, it wouldn't make sense that all the employees are using one shared password, hence each and every client will create their own Wi-Fi passwords to connect to the Wireless Router via the RADIUS Server.



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WPA ENTERPRISE CONNECTION

- At this point the Clients need to be available in the RADIUS Server and Authentication will automatically take place.



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WIRELESS SECURITY STANDARDS (CONTINUES…)

5. **Wireless LAN Security (IEEE 802.11i)**

- Finalized in 2004 and introduced the **Robust Security Network (RSN)** concept.
- It's **features** included;

i. AES-CCMP

ii. 802.1X authentication

iii. Key management, and data integrity

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WIRELESS SECURITY STANDARDS (CONTINUES…)

6. **Port-Based Network Access Control (IEEE 802.1X)**

- Provides a framework for authenticating and authorizing devices on a network.
- It's **components** include;

i. **Supplicant** - The client device requesting access

ii. **Authenticator** – Network device (e.g., Access Point or Switch)

iii. **Authentication Server** - Usually a RADIUS server

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WIRELESS SECURITY STANDARDS (CONTINUES…)

Extensible Authentication Protocol (EAP) Variants Used in IEEE 802.1X

i. **EAP-Transport Layer Security (TLS)** – Uses client and server certificates

ii. **Protected EAP (PEAP)** – Encapsulates EAP inside a TLS tunnel

iii. **EAP-Tunnelled Transport Layer Security (TTLS)**

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4-WAY HANDSHAKE

- Is a process used in wireless security protocols, particularly Wi-Fi Protected Access protocols, to establish a secure connection between a Wi-Fi client (like your laptop or phone) and a Wi-Fi access point (router).
- Its **purpose** is to;

i. To confirm both the client and access point know the Pre-Shared Key (PSK) or use 802.1X authentication.

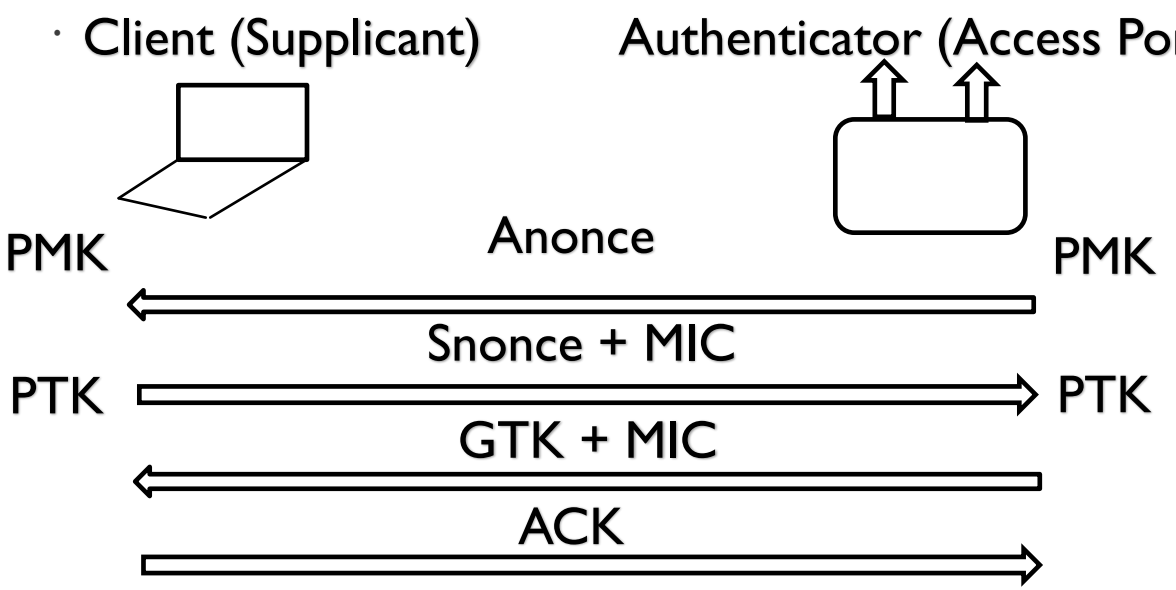
ii. To generate and exchange encryption keys (called Pairwise Transient Keys or PTKs).

iii. To protect against replay and man-in-the-middle attacks.

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4-WAY HANDSHAKE (CONTINUES…)

OPERATION



Acronyms

i. Pre-Master Key (PMK)
ii. Authenticator nonce (Anonce)
iii. Pairwise Transient Key (PTK)
iv. Supplicant nonce (Snonce)
v. Massage Integrity Check (MIC)
vi. Group Temporal Key (GTK)
vii. Acknowledgment (ACK)

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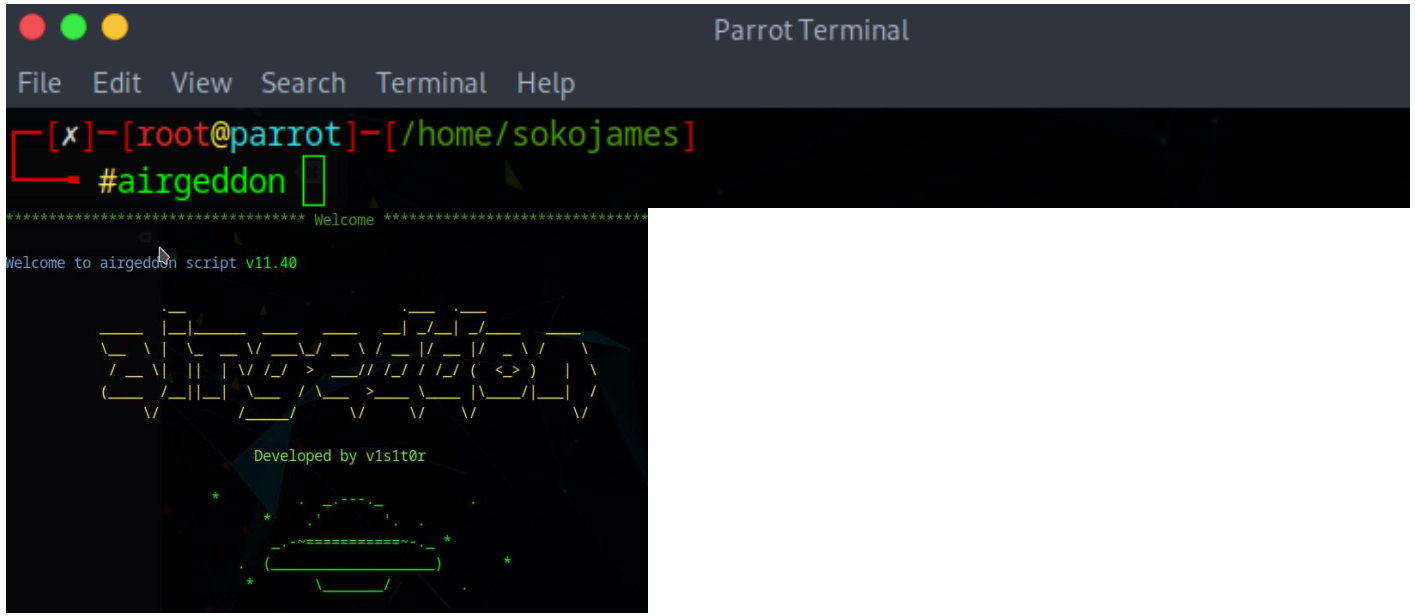
PRACTICAL LAB

AIRCRACK-NG SUITE

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AIRGEDDON

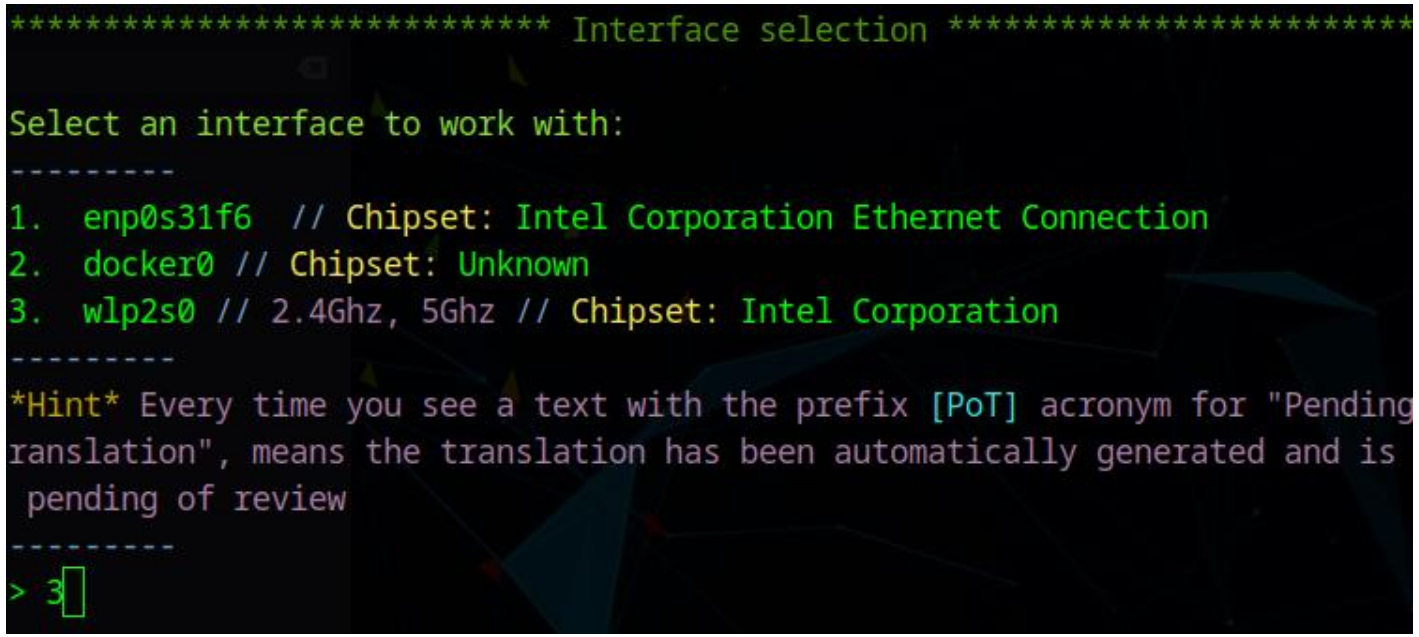
- Open your Linux terminal as root (**sudo su**) and type **airgeddon** if its already installed, if not sudo **apt-get install airgeddon**

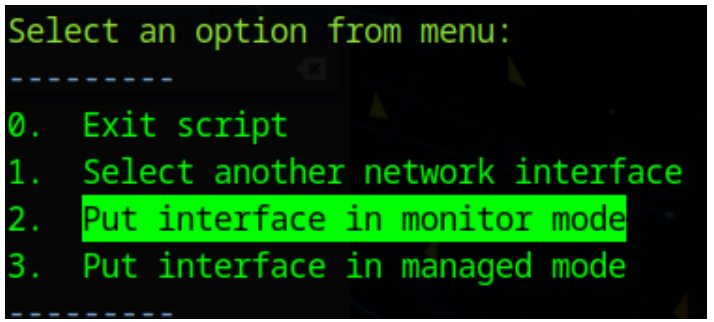


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MONITOR MODE

- Select the Wireless Interface and change the mode from Managed Mode to **Monitor Mode**.

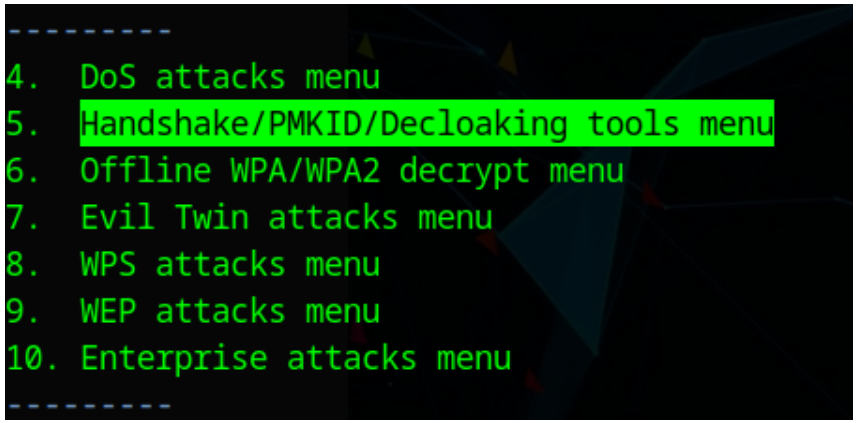


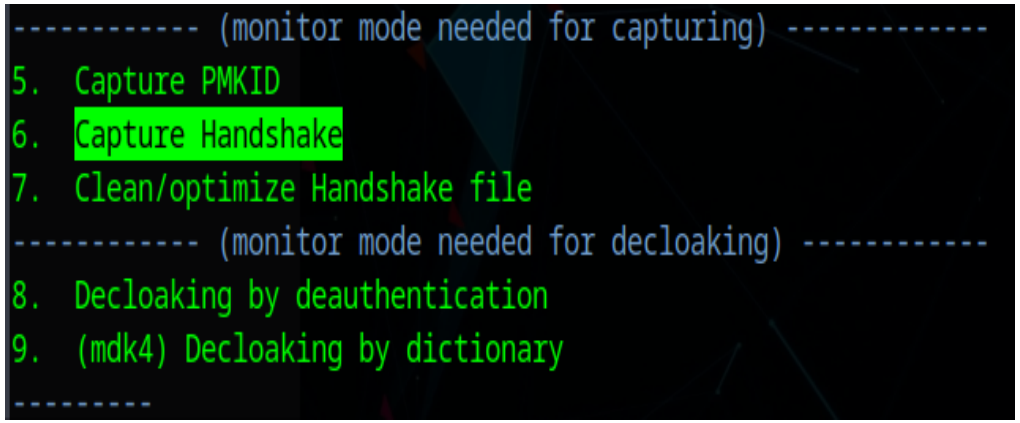


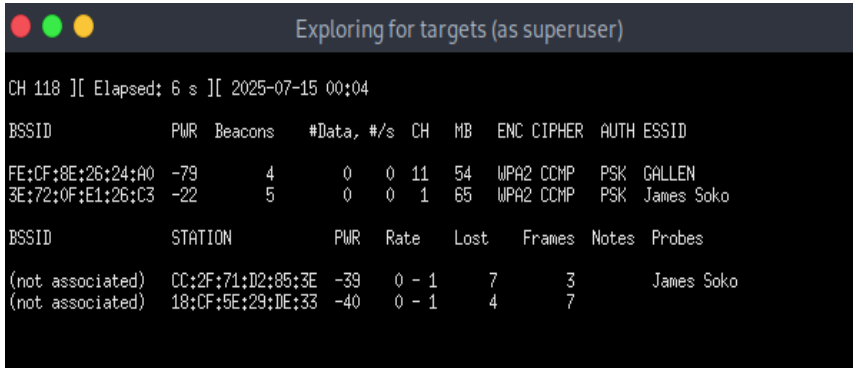
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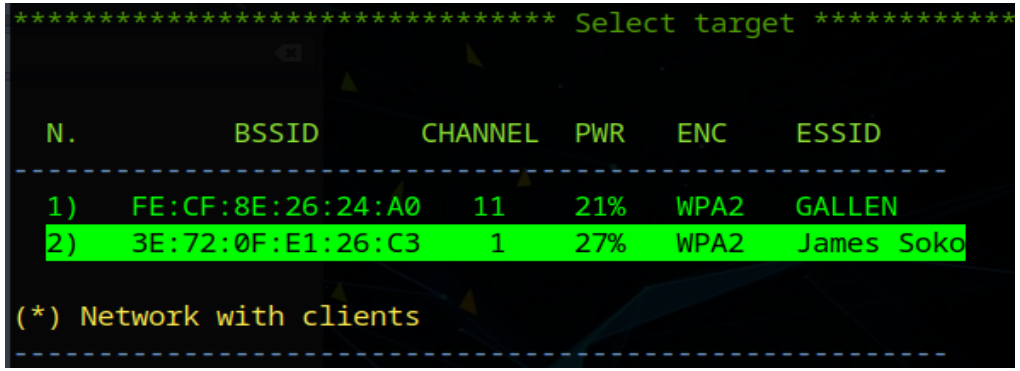
HANDSHAKE

- Establish a Handshake.





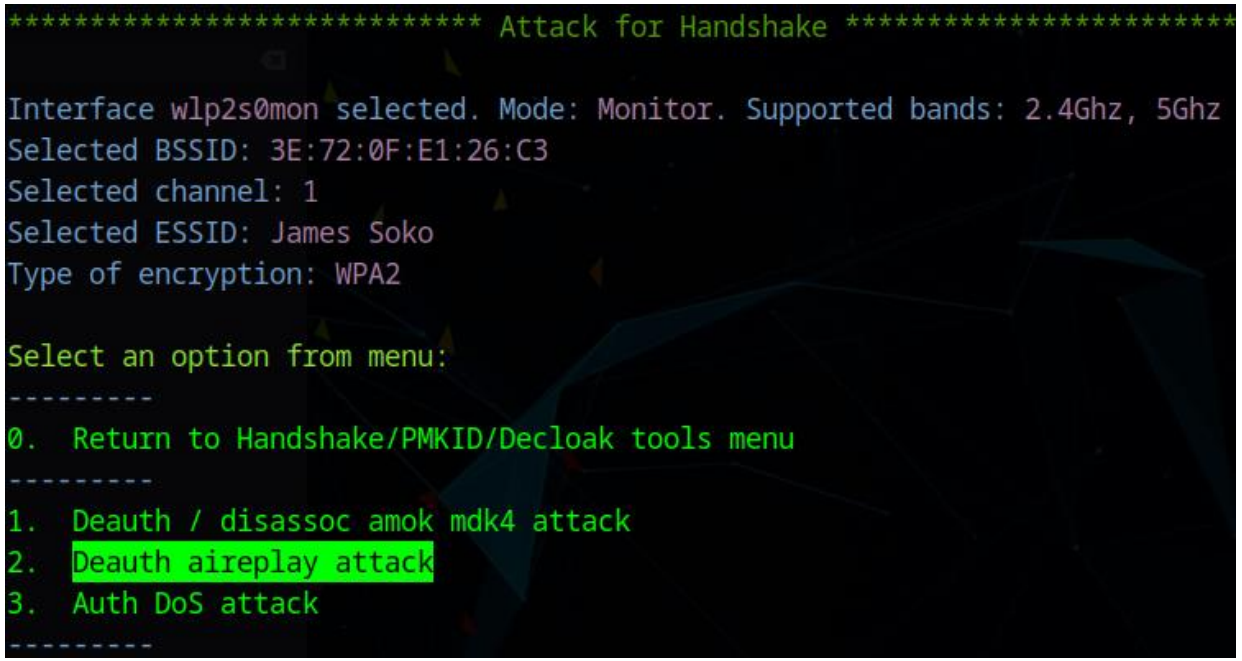




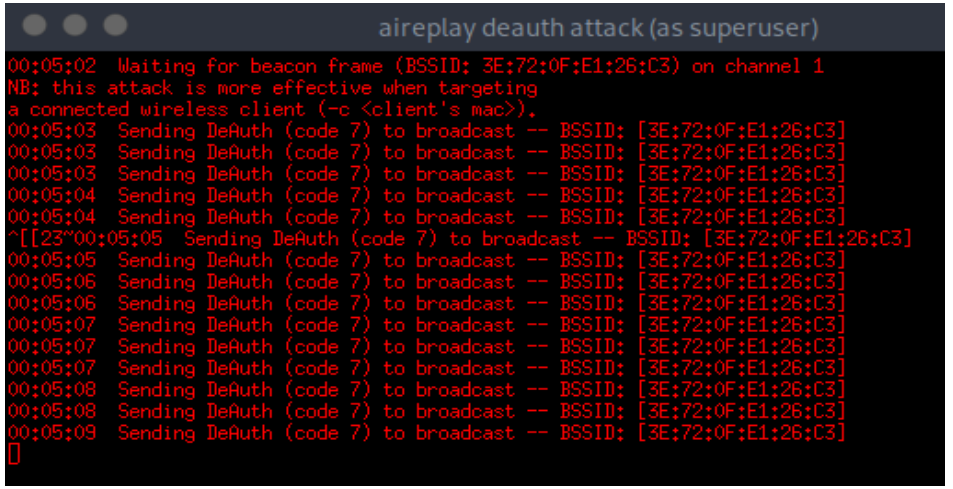
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DE-AUTHENTICATION

- **NOTE:** Someone must be connected to the target Wireless Network (i.e., James Soko) for the de-authentication to be a success.



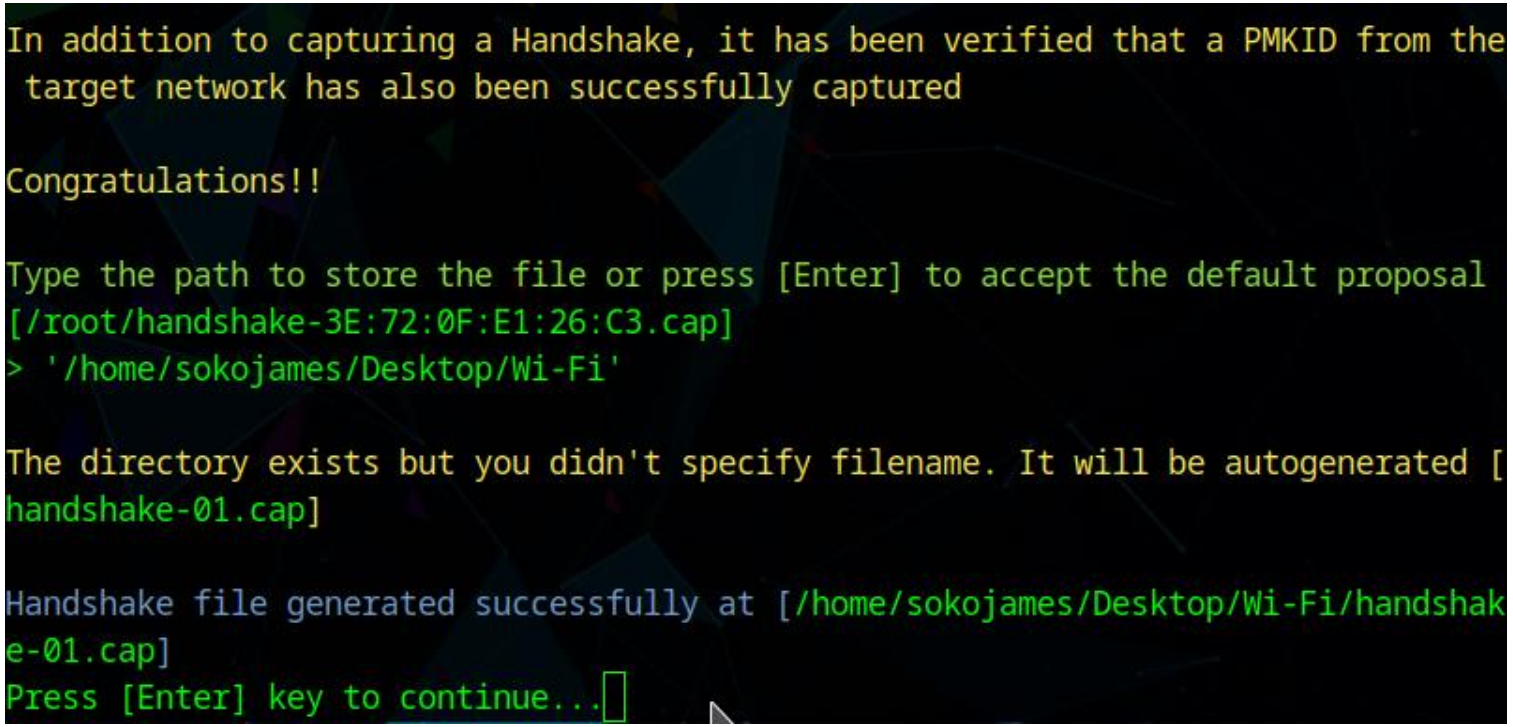


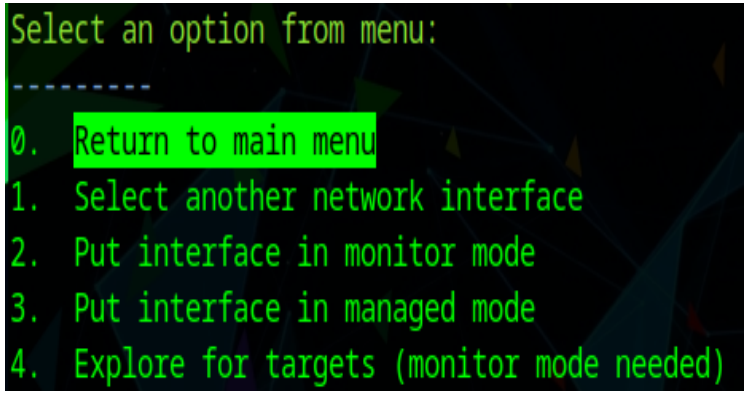


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SUCCESSFUL HANDSHAKE

- Make sure that the handshake and the de-authentication are a success.

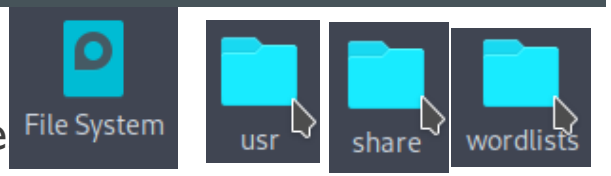




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OFFLINE PASSWORD DECRYPTION

- **NOTE:** We are using a **wordlist** to launch this attack and you can utilize them in the




```
---------
4.  DoS attacks menu
5.  Handshake/PMKID/Decloaking tools menu
6.  Offline WPA/WPA2 decrypt menu
7.  Evil Twin attacks menu
8.  WPS attacks menu
9.  WEP attacks menu
10. Enterprise attacks menu
---------
```

```
Select an option from menu:
----------
0.  Return to main menu
1.  Personal
2.  Enterprise
----------
```

```
Select an option from menu:
---------
0.  Return to offline WPA/WPA2 decrypt menu
-------------- (aircrack CPU, non GPU attacks) ---------------
1.  (aircrack) Dictionary attack against Handshake/PMKID capture file
2.  (aircrack + crunch) Bruteforce attack against Handshake/PMKID capture file
```

```
You already have selected a capture file during this session [/home/sokojames/De
sktop/Wi-Fi/handshake-01.cap]

Do you want to use this already selected capture file? [Y/n]
> Y

You already have selected a BSSID during this session and is present in capture
file [3E:72:0F:E1:26:C3]

Do you want to use this already selected BSSID? [Y/n]
> Y
```

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WI-FI PASSWORD CRACKING

- Will utilize **rockyou** wordlist by unzipping it (drag it to the Desktop and unzip it) because it is the biggest wordlist in Linux (139.9MBs) and there is a possibility we can crack the Wi-Fi (James Soko) password.




```
Enter the path of a dictionary file:
> '/home/sokojames/Desktop/rockyou.txt'
```

```
                    Aircrack-ng 1.7

 [00:00:03] 12224/10303727 keys tested (3746.78 k/s)

 Time left: 45 minutes, 46 seconds                          0.12%

                    KEY FOUND! [ 123456789 ]


 Master Key     : DF 09 8E F3 22 4F CD 52 A8 E3 23 9A 70 C3 AE 1A
                  B2 71 99 4F A9 D0 74 E6 27 3E 17 40 D5 96 DB DF

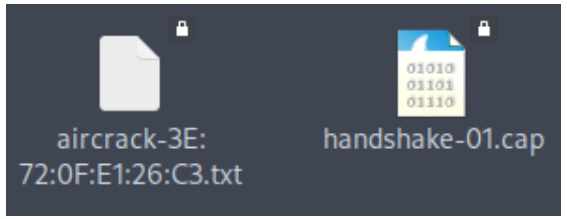
 Transient Key  : EB B1 7A 83 85 EB 5B 4F D6 92 7E 7F 88 7E 0E EE
                  36 85 1E D2 6C DB D0 F0 FA A9 4C C6 79 AE C1 1F
                  22 46 8B 84 1A 39 42 06 6B 3D 3C E1 22 33 DB 41
                  63 55 6A 00 EA 27 07 34 DD 47 7D C2 91 39 B6 7E

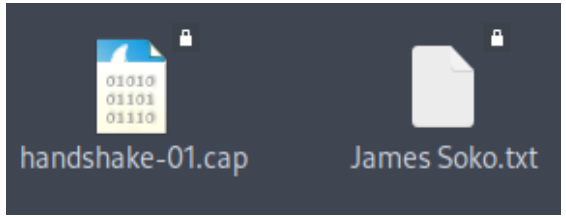
 EAPOL HMAC     : B2 1C CA F9 5C D1 6D 6D BC B4 A4 3D 18 08 A0 48
```

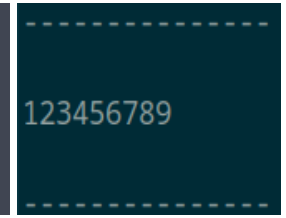
```
Congratulations!! It seems the key has been decrypted

Do you want to save the trophy file with the decrypted password? [Y/n]
> Y

Type the path to store the file or press [Enter] to accept the default proposal
[/root/aircrack-3E:72:0F:E1:26:C3.txt]
> '/home/sokojames/Desktop/Wi-Fi'
```







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